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motion-base: A Ground Motion Database

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Software

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Summary

`motion-base` is a ground motion database designed to provide a unified entry point for accessing and processing ground motion records from various national databases. It includes a web interface and a Python client for programmatic access. The main functionalities include parsing and indexing ground motion records, allowing searches based on different criteria, and providing ad hoc processing such as up-/down-sampling, normalization, filtering, and computation of response spectra. This software is particularly useful for researchers and engineers in earthquake engineering and seismology, offering a streamlined and standardized approach to handle ground motion data efficiently.

Statement of need

The PEER Ground Motion Database (USA) offers a simple search interface that allows users to search for ground motion records based on a few parameters such as magnitude, fault type, etc. The search has a limit of 100 records per query and the records can be downloaded in a plain text format.

The NIED (Japan) also provides a database with a similar search interface. After logging in, one can download records in a plain text format.

The NZSM (New Zealand) provides public access to a complete collection of their strong motion data. But there is no search interface that can be used to filter records based on certain criteria.

The plain text formats used by these databases are not identical, each has slight differences compared to the others. The raw data acquired from these databases are not machine-readable and require manual processing to extract the necessary information.

This makes it difficult for researchers and engineers to perform large-scale analyses on ground motion records from different databases. It would be beneficial to have a FAIR (Wilkinson et al., 2016) facility that provides a unified platform for accessing and processing ground motion records for various applications in earthquake engineering and seismology.

Overview

`motion-base` is designed to address this need. The raw data obtained from different databases can be fed to the parsers, which will extract the metadata and the time series records and store them in a standardized JSON-based format. The records are indexed and stored in MongoDB that offers flexible querying capabilities.

The parsed records can be further processed, common operations include

1. up-/down-sampling and filtering using digital signal processing techniques,
2. computing frequency spectra using the Fourier transform,

3. computing response spectra using the algorithm by Lee (1990).

Compared to the aforementioned databases, in which response spectra are pre-computed with fixed parameters, all these operations are performed on the raw data on-the-fly with `motion-base`, making it easy to perform ad-hoc analyses on the records. Researchers and engineers can flexibly adjust the parameters of the processing steps to suit their needs.

Figure 1 shows the landing page of the web interface.

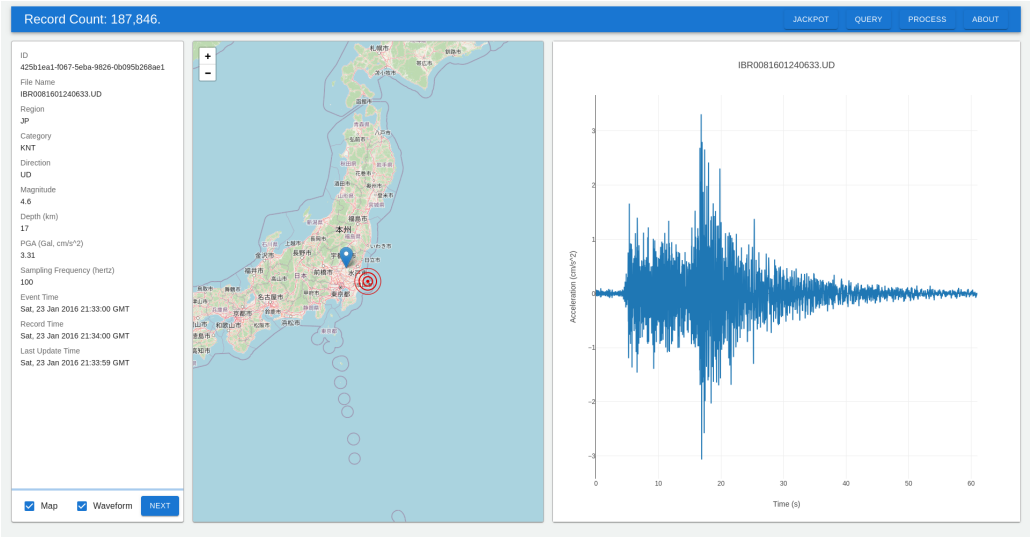

Figure 1: The landing page of the web interface.

Deployment

`motion-base` can be deployed on UNIX-like systems. For the purpose of illustration, the following commands will create an example deployment on a local machine.

```
git clone --depth 1 --branch master https://github.com/TLCFEM/motion-base.git
bash motion-base/scripts/example.sh
```

References

Lee, V. W. (1990). Efficient algorithm for computing displacement, velocity and acceleration responses of an oscillator to arbitrary ground motion. *Soil Dynamics and Earthquake Engineering*, *9*(6), 288–300. https://doi.org/10.1016/s0267-7261(05)80015-6

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